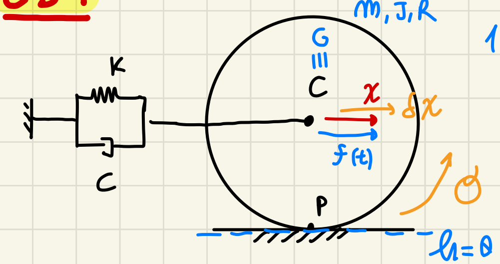
ESE 1 1 DOF, LAGRANGE

es 1



1 DOF linear system
(similar to linear oscillator
BUT with a Roll without sliding disk)
↓
Just one between x_c, θ is independent, we can choice which one to use!

(we prefer to use ROTATION usually) → HERE we decide to use x_c as INDIPENDENT cohordinate
calling it x ⇐

Lagrange
$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{x}}\right) - \frac{\partial T}{\partial x} + \frac{\partial D}{\partial \dot{x}} + \frac{\partial V}{\partial x} = \frac{\delta W_{NC}}{\delta x}$$
($f(t)$ acting on disk center)

RIGID Body (DISK) → with uniform mass distribution, center of mass $G \equiv C$
(velocity vector respect point C!) as center of mass
we have both LINEAR + ANGULAR velocity

KINETIC ENERGY

$$T = \frac{1}{2} m |\vec{v}_G|^2 + \frac{1}{2} J_G |\vec{\omega}|^2$$
↑
KÖNIGS theorem → total T as summ of two terms

{ m and J_G are given parameters! } PROBLEM DATA
↑
simple formula to evaluate $J = \ldots m$

{ for simple disk $J_G = \frac{mR^2}{2}$ } known!

Using $\dot{x}$ as velocity of C
$\dot{\theta}$ as Rotation (angular velocity)
$$T = \frac{1}{2} m \dot{x}^2 + \frac{1}{2} J \dot{\theta}^2$$

BUT as we know we can say: $\| \dot{\theta} = -\dot{x}/R \|$ because of ROLLING without sliding and $\vec{v}_P = 0$!

$$T = \frac{1}{2} \underbrace{\left(m + J/R^2\right)}_{\bar{m}} \dot{x}^2 = \frac{1}{2} \bar{m} \dot{x}^2$$

final kinetic energy expression with similar express of mass oscillator ←
here we have a corresponding mass $\bar{m}$ of the system

do measure verification
$[J/R^2] = [Kg]$ MASS

$T = \frac{1}{2} \bar{m} \dot{x}^2$ $[J] = [N \cdot m] = [kg\, m^2/s^2]$ ENERGY — check always unit of measure!

DISSIPATION function

$$D = \frac{1}{2} c \underbrace{\dot{\Delta l}}^2 = \frac{1}{2} c \dot{x}^2$$

already function of x JUST ready for lagrange

dashbod elongation → easy to evaluate! it is the moovement of $c \to x$ $\dot{\Delta l} = \dot{x}$

potential energy

$$V = \underset{\uparrow}{V_{ee}} + V_g = \underbrace{\frac{1}{2} k \Delta l^2} + mg\, h =$$

linear massless spring — elastic pot energy — weight — height of center of mass respect Reference → choose reference on the guide of rolling

$$= \frac{1}{2} K x^2 + \cancel{mgR} \quad \frac{\partial}{\partial x} \to 0$$

constant — discarded a priori, NOT usefule!

Virtual WORK

$$\delta W = f(t) \cdot \underset{\sim}{\delta x}$$

virtual dispacement of c when infinitesimal variation of x

JUST modulus multiplication as we see $f, \delta x$ has same direction!

⇓ NOW we JUST compute LAGRANGE

$$\frac{\partial T}{\partial \dot{x}} = \frac{\partial}{\partial \dot{x}}\left(\frac{1}{2}\bar{m}\dot{x}^2\right) = \bar{m}\dot{x} \to \frac{d}{dt}(\bar{m}\dot{x}) = \bar{m}\ddot{x}$$

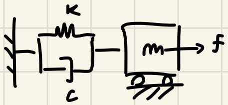
↑ constant

kinematically linear system so $\bar{m}$ is const, does NOT depend on x! generalized mass

LAGRANGE

$$\bar{m}\ddot{x} - 0 + c\dot{x} + Kx = f(t)$$

EQUATION of MOTION { same of the linear mass slider }



↖ here with a generalized mass $\bar{m}$

IF we use a DISK ⇐ non hollow with uniform density

$$J = \frac{m}{2} R^2 \quad \text{so:} \quad \bar{m} = \left(m + \frac{m}{2}\frac{\cancel{R^2}}{\cancel{R^2}}\right) = \frac{3}{2} m$$

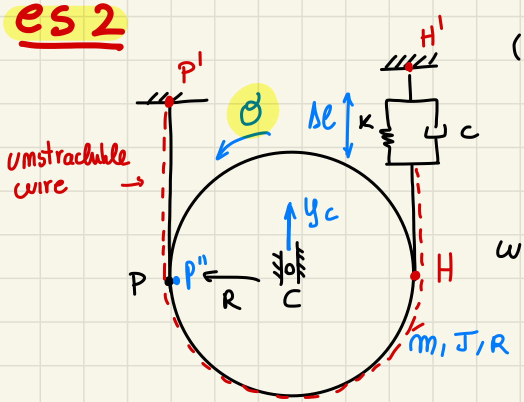
general mass. you have more mass in the DISK respect the Real one!

(this is why in bycicly we TRY to save mass on wheels!)

NOTICE: some issue are very relevant on REAL LIFE application

remove wheel mass to have a less inertia motion
→ the bike acceleration has an influence of the BIKE greater than it's real mass... with HIGHER RATIO even because of the displacement an constructive feature

es 2



(RIGID disk) uniform mass

constrain on C, so disk can only ROTATE and moove along y!

WIRE around the DISK and contact to SPRING + DUMP letting the disk moove (δe) elongation

a WIRE in TENSION is a constrain (−1 DOF) and (−1 DOF due to CART on C)
D.O.C D.O.C

HH' NOT constrains because SPRING let moove

3DOF disk − 1 CART − 1 WIRE → 1 DOF

C can moove only VERTICALLY y_c and ROTATE disk θ

y_c, θ related because 1 DOF ONLY! ←

because UNSTREACHABLE WIRE $v_p = v_p' = 0$!

$\dot{y}_P = 0$ same velocity of P'

DISK cannot SLIDE on P (like on a BELT) enough FRICTION to avoid sliding

P'' ∈ DISK
P ∈ WIRE → $\dot{y}_{P''} = \dot{y}_P = 0$!

so for KINEMATICS for Rigid bodies, RIVALS... Theorem

$$\dot{y}_{P''} = \dot{y}_c - R\dot{\theta} = 0$$

$$\Downarrow$$

$$\dot{y}_c = R\dot{\theta}$$ → INTEGRATING you relate y_c, θ

y_c, θ are dependent in between

we choice θ as INDIP. variable

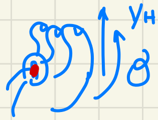
$$y_c = R\theta + \cancel{const}$$

0 with PROPER origin choice

We need to know H velocity, on opposite DISK SIDE... because H motion determines the δe of SPRING because CART constrains NO points move along x, only along y VERTICAL MOTION

$$\dot y_H = \dot y_C + R\dot\theta = 2R\dot\theta$$ Double velocity RESPECT the $\dot y_C$ velocity! $\Rightarrow$ $y_H = 2R\theta$ ~~+const~~ $\leftarrow$ 0 proper origin choice

$\uparrow$ {RIVALS THEOREM}



same direction

(as seen on MODELLISTICA)

P ≡ C.I.R center of Instantaneous Rotation

single point With 0 speed ! → from P you can express any POINT velocity as vector product $\dot\theta \cdot$ distance point–CIR

perpendicular velocity to direction with P

$\dot\theta \cdot$ distance (POINT, C.I.R(P)) directing according to (Right HAND Rule)

find C.I.R position ease the evaluation of VELOCITY

(in Rolling without sliding the CIR is on P of contact)

NOW we have to evaluate the energetic quantities

to evaluate
$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot\theta}\right) - \frac{\partial T}{\partial \theta} + \frac{\partial D}{\partial \dot\theta} + \frac{\partial V}{\partial \theta} = \frac{\delta W_{NC}}{\delta \theta}$$

kinetic energy

$$T = \frac{1}{2} m \dot y_c^2 + \frac{1}{2} J \dot\theta^2 = \frac{1}{2}\underbrace{(mR^2 + J)}_{\bar J}\dot\theta^2 = \frac{1}{2}\bar J \dot\theta^2 \ [J]$$

Generalized mass moment of inertia of body

dissipation function

$$D = \frac{1}{2} c (\Delta\dot\ell)^2 = \frac{1}{2} c\, 4R^2 \dot\theta^2$$

IT Is easy to see $\Delta\dot\ell = -\dot y_H$

considering $\Delta\ell > 0$ When it is a damper elongation y_H moves up $y_H > 0$ so $\Delta\ell$ is opposite !

$$\Delta\dot\ell = -\dot y_H = -2R\dot\theta$$

potential energy

height of center of mass respect reference 0

$$V = V_{el} + V_g = \frac{1}{2} k \Delta\ell^2 + m g y_c =$$

$$\Delta\ell = -2R\theta + \Delta\ell_0$$

here the integration constant MUST be considered! $\Rightarrow$ $\Delta\ell_0$ ELONGATION of spring When $\theta = 0$ value

$\vartheta_0 = 0$ or a general ϑ Reference 0

$\partial/\partial\vartheta$ goes to 0, CONSTANT!

$$\rightarrow V = \frac{1}{2} K \left(4R^2\vartheta^2 - 4R\Delta l_0 \vartheta + \cancel{\Delta l_0^2}\right) + mgR\vartheta$$

function of ϑ, CANNOT be neglected a priori

here NO FORCE ACTING $\delta W_{NC} = 0$

KINEMATICALLY LINEAR System!

T does NOT depend on ϑ

$\Downarrow$ LAGRANGE: (deriving the terms)

NO time dependent forces acting!

$$\underbrace{\overbrace{(mR^2 + J)}^{\bar{J}}\ddot{\vartheta} - 0 + \overbrace{4cR^2}^{\bar{C}}\dot{\vartheta} + \overbrace{4KR^2}^{\bar{K}}\vartheta}_{\text{this is usual form of eq. of MOTION}} - 2RK\Delta l_0 + mgR = 0 \quad (*)$$

Related to STATIC forces acting on the system... (Weight) and (PRE-LOAD of SPRING)

$\downarrow$ elongation of SPRING in its equilibrium condition

usually this terms can be set to 0 depending on the ϑ_0 choice of the reference ϑ 0 value

2 options:

1. $\vartheta = 0$ where $\Delta l_0 = 0$, so where there is NO PRELOAD, spring NON extended for $\vartheta = 0 \rightarrow \Delta l_0 = 0$ while $mgR \neq 0$

$\downarrow$

$$\bar{J}\ddot{\vartheta} + \bar{C}\dot{\vartheta} + \bar{K}\vartheta = -mgR$$

costant force due to weight

2. set $\vartheta = 0$ where we have EQULIBRIUM position of the system!

SO.. IF $\vartheta = 0$ @ equilibrium

$\vartheta = 0 = \text{const}$ MUST be solution of motion equation $(*)$

problem solution

$\Downarrow$

$$\dot{\vartheta} = 0, \ \ddot{\vartheta} = 0 \quad \text{because const}$$

meaning $-2K\cancel{R}\Delta l_0 + mg\cancel{R} = 0$

PRE-LOAD cancel the DISK weight effect!

always possible for KINEMATICALLY linear system!

(IN THIS lucky case, SET $g = 0$ where equilibrium, means weight acting on system disappear because cancelled by PRE-LOAD of SPRING)

so you came back to a general $m\ddot{x} + c\dot{x} + kx = 0$!

eq. of motion

USUAL SOLUTION